# Discussion Section 5
# STAT 132 , Winter 2025

1. Find the mode of the $\chi^2_m$ distribution for $m = 1, 2, \ldots$

2. Suppose that a point $(X, Y)$ is to be chosen at random in the xy-plane, where $X$ and $Y$ are independent random variables and each has the standard normal distribution. If a circle is drawn in the xy-plane with its center at the origin, what is the radius of the smallest circle that can be chosen in order for there to be probability 0.99 that the point $(X, Y)$ will lie inside the circle?

3. Suppose that six random variables $X_1, \ldots, X_6$ form a random sample from the standard normal distribution, and let
$$Y = (X_1 + X_2 + X_3)^2 + (X_4 + X_5 + X_6)^2.$$
Determine a value of $c$ such that the random variable $cY$ will have a $\chi^2$ distribution.

4. Assume that $X_1, \ldots, X_n$ form a random sample from the normal distribution with mean $\mu$ and variance $\sigma^2$. Show that $\widehat{\sigma^2}$ has the gamma distribution with parameters $(n1)/2$ and $n/(2\sigma^2)$.

5. Suppose that $X_1, \ldots, X_n$ form a random sample from the normal distribution with mean $\mu$ and variance $\sigma^2$. Assuming that the sample size $n = 16$, determine the values of the following probabilities:

   (a)
   $$\mathbb{P}\left(\frac{\sigma^2}{2} \leq \frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)^2 \leq 2\sigma^2\right)$$

   (b)
   $$\mathbb{P}\left(\frac{\sigma^2}{2} \leq \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2 \leq 2\sigma^2\right)$$